

Data Warehousing and Modeling

1 What is a Data Warehouse?

W. H. Inmon's Definition

"A data warehouse is a **subject-oriented**, **integrated**, **time-variant**, and **nonvolatile** collection of data in support of management's decision-making process."

It serves as a decision support database maintained separately from the organization's operational databases, providing a solid platform of consolidated, historical data for analysis.

1.1 The Four Characteristics

1. Subject-Oriented:

- Organized around major subjects, such as **customer**, **product**, **sales**.
- Focuses on modeling and analysis for decision makers, not on daily operations or transaction processing.
- Provides a simple and concise view by excluding data that are not useful in the decision support process.

2. Integrated:

- Constructed by integrating multiple, heterogeneous data sources (relational databases, flat files, on-line transaction records).
- Data cleaning and integration techniques are applied to ensure consistency in naming conventions, encoding structures, and attribute measures.
- *Example:* Reconciling hotel prices that might have different currency, tax, or breakfast coverage across sources.

3. Time-Variant:

- The time horizon is significantly longer than operational systems (e.g., **5–10 years** vs. current value data).
- Every key structure contains an element of time, explicitly or implicitly.
- Operational data may or may not contain a "time element" in its key, but warehouse data always does to provide a historical perspective.

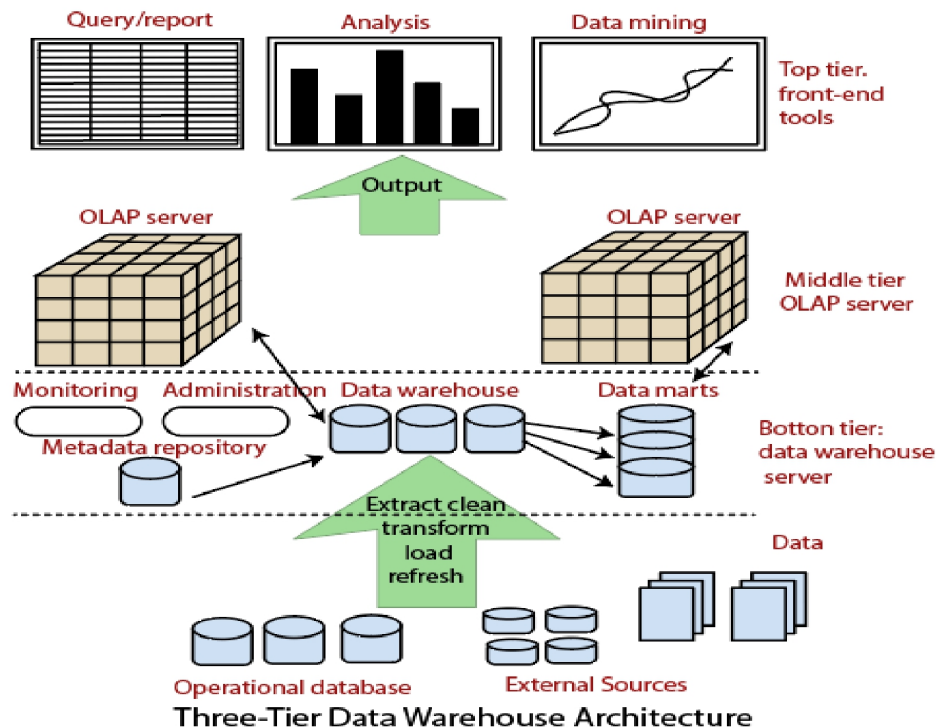
4. Non-Volatile:

- A physically separate store of data transformed from the operational environment.

- Operational updates do not occur here; it does not require transaction processing, recovery, or concurrency control.
- Requires only two operations in data accessing: **initial loading** and **access**.

2 Three-Tier Data Warehouse Architecture

A data warehouse typically adopts a three-tier architecture that separates data sources, the warehouse server, and front-end analytical tools.



18

2.1 Data Warehouse vs. Heterogeneous DBMS

- **Traditional Heterogeneous DB Integration (Query-Driven):**
 - Build **wrappers/mediators** on top of heterogeneous databases.
 - When a query is posed to a client site, a **meta-dictionary** is used to translate it into queries appropriate for individual heterogeneous sites involved.
 - Results are then integrated into a global answer set.
 - **Drawbacks:** Complex information filtering, high computation cost, and competes for local source resources.
- **Data Warehouse (Update-Driven):**
 - Information from heterogeneous sources is **integrated in advance** and stored in the warehouse for direct query and analysis.
 - Provides high performance as complex filtering and integration are pre-processed.

2.2 OLTP vs. OLAP

Data Warehouses perform **OLAP** (On-line Analytical Processing), while traditional RDBMS perform **OLTP** (On-line Transaction Processing).

- **OLTP:** Major task of traditional relational DBMS. Focuses on day-to-day operations: purchasing, inventory, banking, manufacturing, payroll, registration, accounting, etc.
- **OLAP:** Major task of data warehouse systems. Focuses on data analysis and decision making.

Distinct Features (OLTP vs. OLAP)

Feature	OLTP	OLAP
User	Clerk, IT professional	Knowledge worker (manager, analyst)
Function	Day-to-day operations	Decision support
Orientation	Customer-oriented	Market-oriented
Data Contents	Current, isolated, detailed	Historical, integrated, consolidated
DB Design	E-R + Application-oriented	Star/Subject-oriented
View	Current, local	Evolutionary, integrated
Access Patterns	Update (Write/Read)	Read-only, complex queries

2.3 Why Separate them?

- **High Performance for Both Systems:**
 - **Operational DBMS:** Tuned for OLTP (fast updates, indexing on keys, concurrency control, and recovery).
 - **Data Warehouse:** Tuned for OLAP (complex multidimensional queries, consolidation, and reporting).
- **Different Functions and Data:**
 - **Historical Perspective:** Operational DBs lack long-term historical data required for decision support.
 - **Data Consolidation:** Decision support requires aggregation and summarization from multiple heterogeneous sources.
 - **Data Quality:** Operational sources often use inconsistent data representations, codes, and formats that must be reconciled (Cleaned/Integrated).

3 Conceptual Modeling: Schemas

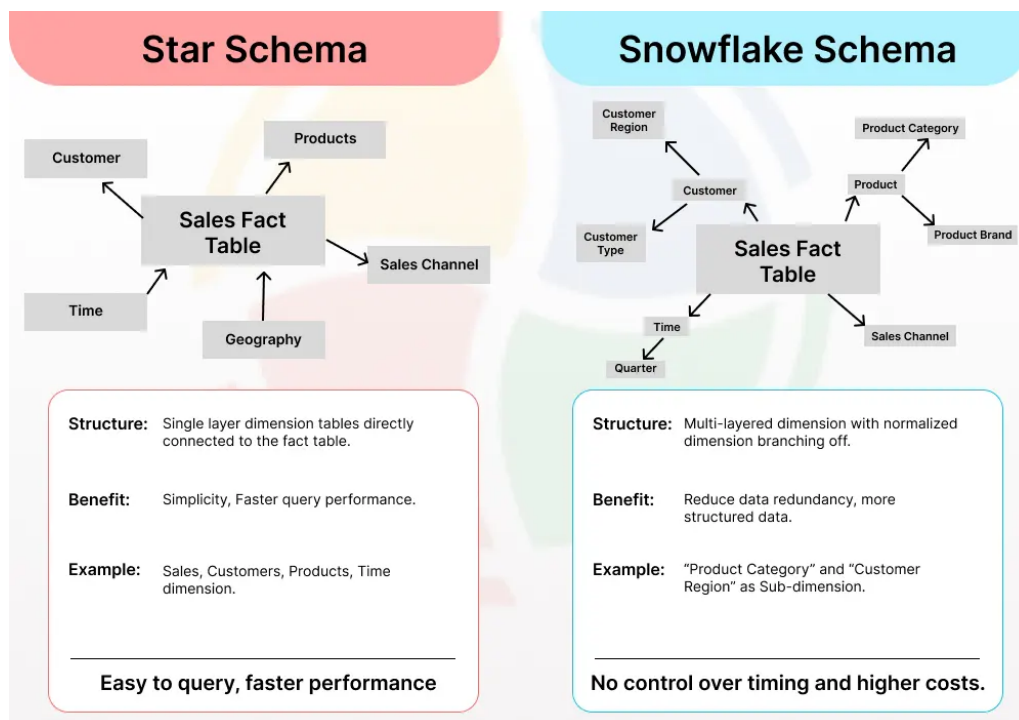
Modeling data warehouses involves ****Dimensions**** (categories) and ****Measures**** (numerical values).

3.1 Star Schema

A single **Fact Table** in the middle is connected to a set of **Dimension Tables**. It is simple and highly efficient for queries.

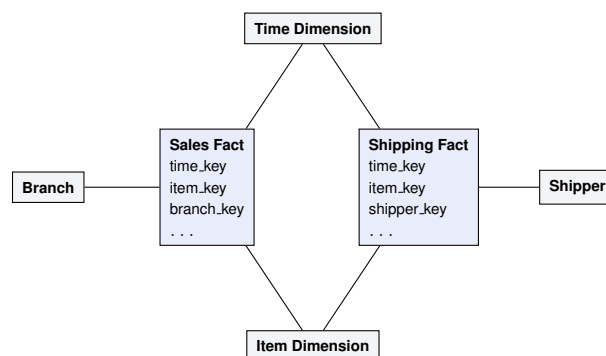
3.2 Snowflake Schema

A refinement of the Star Schema where functional hierarchies are **normalized** into smaller dimension tables, reducing redundancy but increasing query complexity.



3.3 Fact Constellation (Galaxy Schema)

Multiple fact tables share dimension tables. It is a collection of stars that provides a unified view across different business processes (e.g., Sales and Shipping).



Data Warehousing transforms raw operational data into a strategic business asset.